Milestone-04 Report
Incident Type Analysis

Domain: Data Visualization

Project: U.S. Natural Disaster Analysis

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1. Objective

The objective of this milestone is to analyze disaster incident types in detail.

It focuses on understanding how disasters are distributed across different regions.

The study uses FEMA disaster declaration data for insights.

It aims to identify frequently occurring disaster categories.

The analysis also explores relationships with assistance programs.

This helps in understanding disaster impact and response patterns.

2. Problem Formulation

This analysis is based on exploratory and comparative techniques.

Categorical data analysis is used to evaluate disaster types.

Key questions guide the analysis process effectively.

The study identifies frequent disaster occurrences.

It evaluates state-wise disaster distributions.

It also examines how assistance programs vary by disaster type.

3. Dataset Preparation

The dataset used is FEMA disaster declarations data.

Relevant columns include incident type, state, and assistance flags.

Data cleaning was performed in earlier milestones.

Missing values and inconsistencies were handled.

The dataset was validated for accuracy and completeness.

4. Incident Type Aggregation

Data aggregation helps summarize disaster occurrences.

Incident types are grouped to calculate total counts.

State-wise grouping reveals regional disaster patterns.

Aggregation with assistance programs shows impact levels.

These groupings simplify complex datasets.

They enable easier visualization and interpretation.

5. Visualizations

Various charts are used to represent disaster data visually.

Bar charts show frequency of disaster types.

Treemaps illustrate proportional distribution effectively.

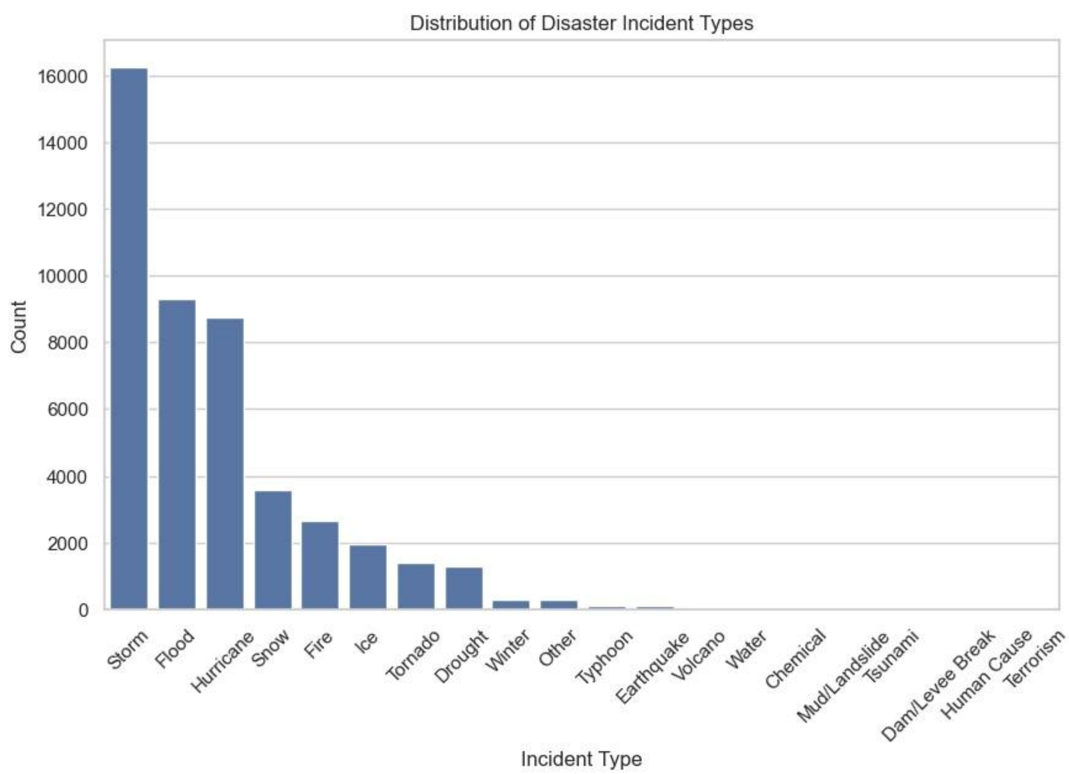
Heatmaps highlight intensity across states.

Stacked charts compare multiple variables.

Visualizations make patterns easier to understand.

5.1 Distribution of Disaster Incident Types

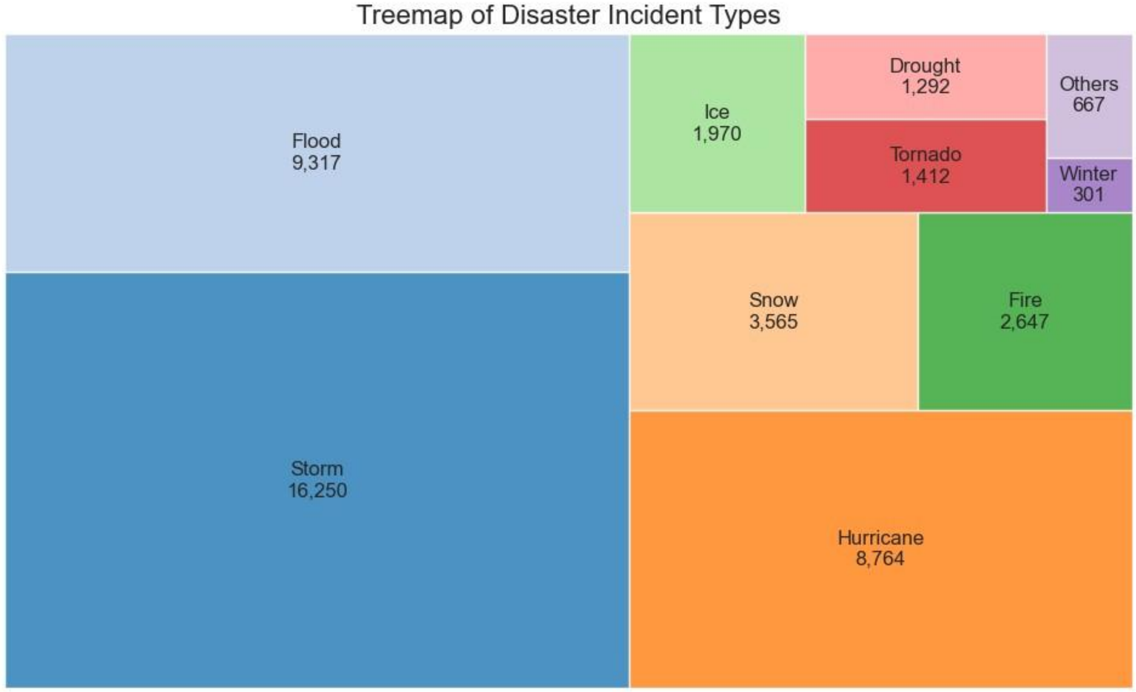
This bar chart displays the total number of disaster declarations for each incident type by aggregating the dataset based on the incident type column.



This bar chart represents the frequency of each disaster type. It shows that certain disaster types occur more frequently than others. Weatherrelated disasters dominate the dataset, indicating their high occurrence rate.

5.2 Tree map of Disaster Incident Types

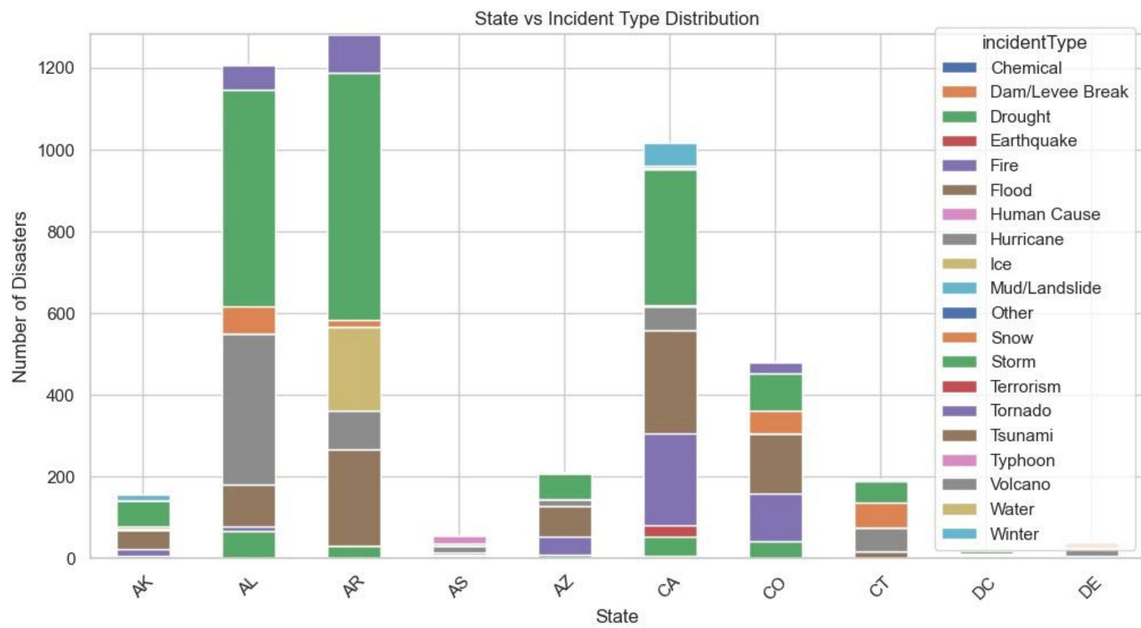
This tree map visualizes the proportion of different disaster types using hierarchical blocks, where each block size represents the frequency of that incident type.



The treemap visualizes disaster types based on their frequency using hierarchical blocks. Larger blocks indicate more frequent disaster types. This visualization provides a quick and effective comparison of disaster composition.

5.3 State vs Incident Type Distribution

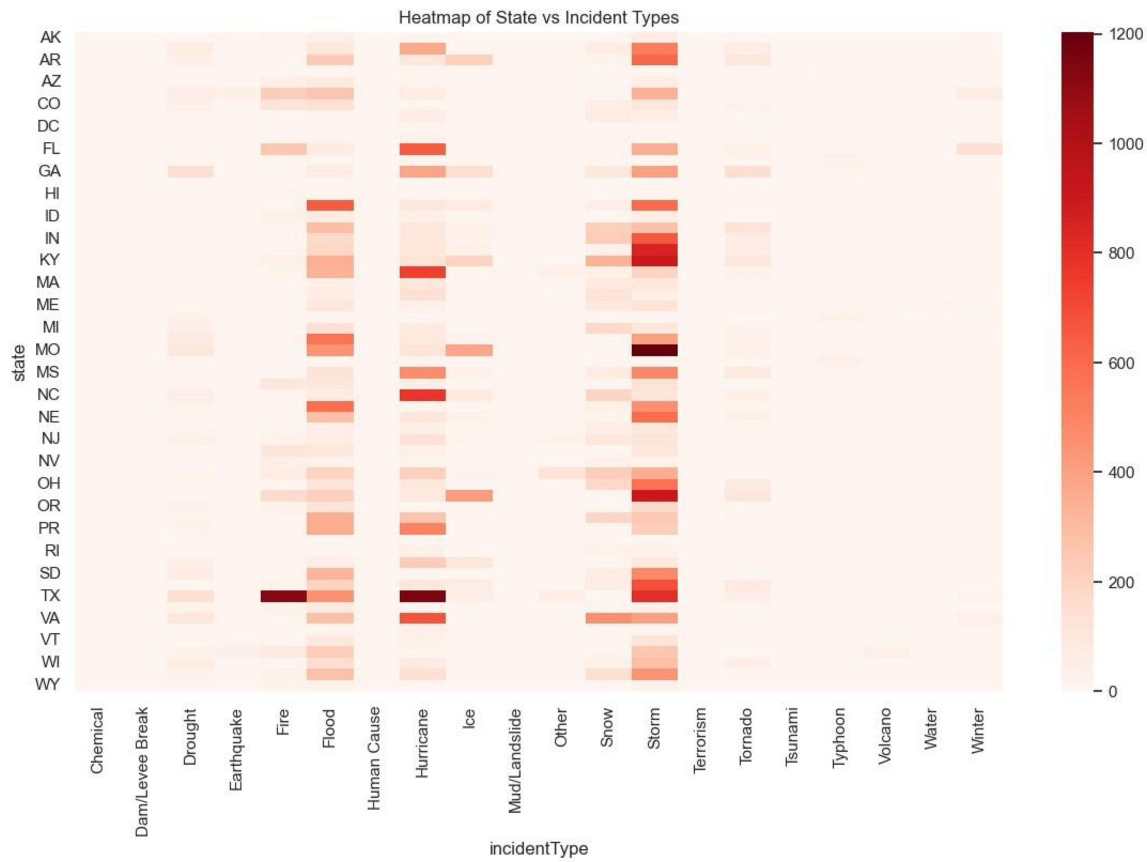
This stacked bar chart represents the distribution of different disaster types across states by grouping the dataset using state and incident type.



This stacked bar chart shows how different disaster types are distributed across states. Different states experience different types of disasters. It helps identify regional patterns and dominant disaster types in specific states.

5.4 Heatmap of Disaster Incident Types by State

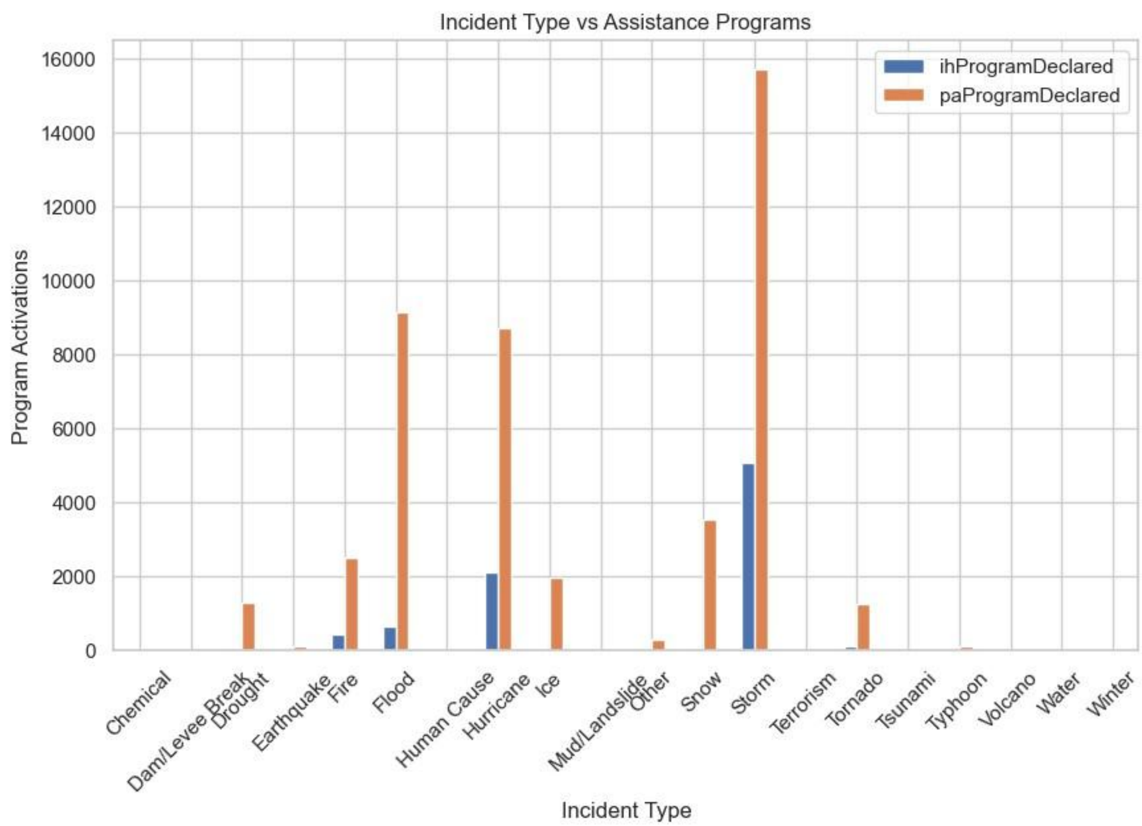
This heatmap shows the frequency of disaster types across states using a matrix format, where rows represent states and columns represent incident types.



The heatmap represents disaster frequency across states and incident types using color intensity. Darker colors indicate higher disaster occurrence. This visualization helps identify disaster hotspots and regional concentration patterns.

5.5 Disaster Assistance Programs by Incident Types

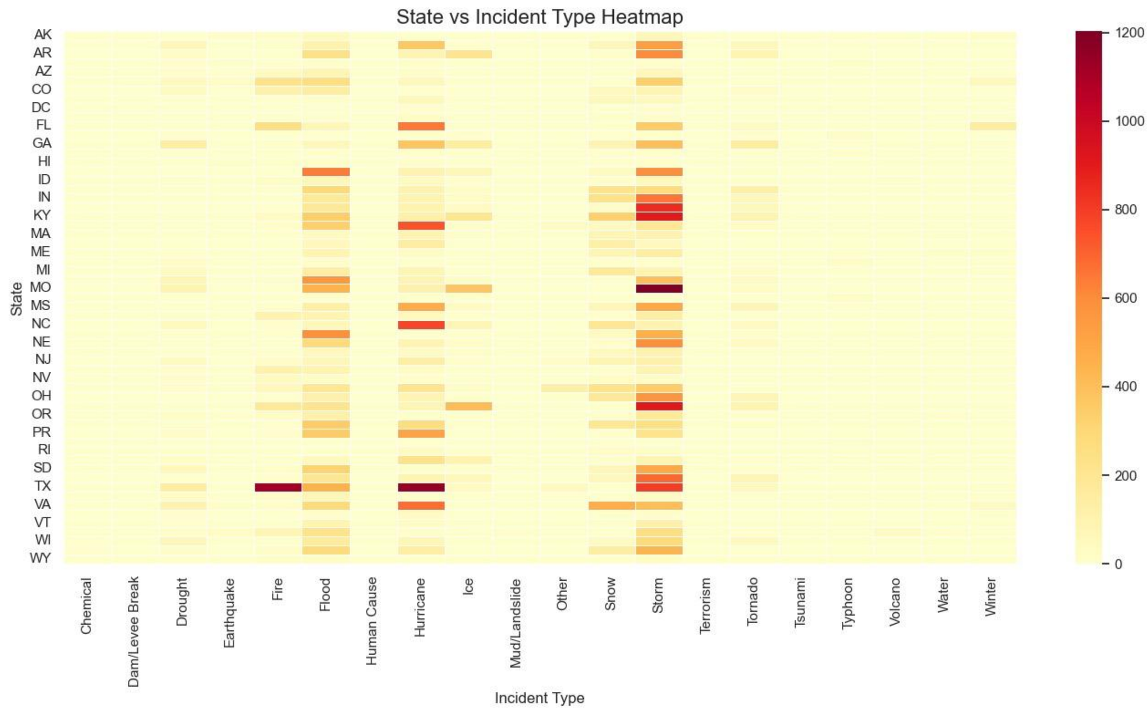
This bar chart illustrates the relationship between disaster types and assistance programs by aggregating ihProgramDeclared and paProgramDeclared for each incident type.



This bar chart shows how disaster types are associated with assistance programs. Certain disaster types require more assistance compared to others. It helps understand the impact and severity of different disasters.

5.6 State vs Incident Type Heatmap

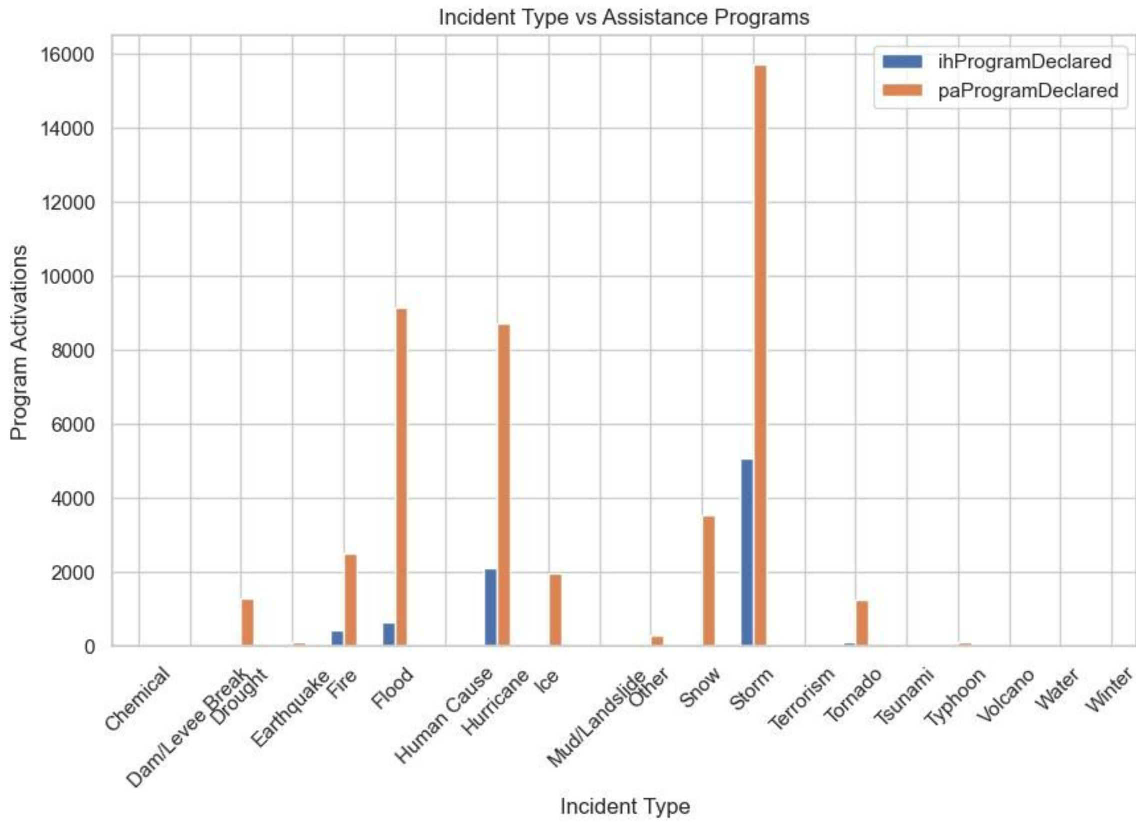
This heatmap provides a detailed visualization of disaster intensity across states and incident types, highlighting variations using color gradients.



This heatmap provides deeper insights into disaster distribution across states and incident types. It highlights intensity and concentration of disasters. This visualization helps identify dominant disaster patterns at the state level.

5.7 Incident Type vs Assistance Programs

This grouped bar chart compares different disaster types with individual and public assistance programs to show how assistance varies by incident type**.**



This visualization compares incident types with individual and public assistance programs. It shows how different disaster types influence assistance requirements. Some disasters require more public assistance, while others impact individuals more directly.

6. Tools and Libraries Used

Pandas is used for data manipulation and preprocessing.

NumPy supports numerical operations efficiently.

Matplotlib is used for basic visualizations.

Seaborn enhances statistical data representation.

Plotly enables interactive and dynamic charts.

These tools together ensure comprehensive analysis.

7. Key Insights

Certain disaster types occur more frequently than others.

Different states show different disaster patterns.

Some regions are prone to specific disasters.

Assistance programs vary depending on disaster severity.

Visual analysis reveals hidden patterns effectively.

Insights help in planning disaster response strategies.

8. Conclusion

This milestone provides a detailed analysis of disaster types.

It highlights patterns across regions and categories.

The relationship between disasters and assistance is explored.

Findings show consistent trends in disaster occurrences.

The analysis supports better understanding of disaster impact.

It forms a base for further advanced analytics.

9. Dashboard Link

The project dashboard is deployed online for interaction.

It provides dynamic visualization of disaster data.

Users can explore trends and patterns easily.

The dashboard enhances accessibility of insights.

It supports decision-making with visual evidence.

Dashboard Link: https://u-s-natural-disaster-dashboard.vercel.app/